

Mounting of the sensor IC requires some special attention. If temperature of a solid surface is to be monitored, the sensor may be fixed to the surface by using metal clamps, or glued directly to the surface with high-temperature epoxy adhesive.

For liquid temperature measurement, the sensor cannot be immersed directly in the liquid, as the liquid may be a conductive type, and in that case, the sensor's leads would be electrically shorted. To solve this problem, the sensor may be mounted inside a sealed-end metal/glass tube. Connecting wires must go to the leads through high-temperature insulating sleeving. The tube may then be dipped into the liquid, or screwed through some threaded hole in the container of the liquid. Although steel gives very high ruggedness to construction, in general, a glass tube would be an ideal choice when temperature of a chemically reactive bath is to be measured.

During construction, special care must be given regarding the choice of some resistors and capacitors. Resistors R1, R2, R24, 25 and 26 must be of highly-stable and low-temperature co-efficient type, to make the units stable against time and ambient temperature variations. Metal-film type may be used for this purpose. VR1 is a 10-kilo-ohm, 40-turn trim potmeter. This should be of highly stable type.

Ground return leads of the sensors must be grounded close to ADC ground lead, otherwise erroneous results may be observed.

Calibration and adjustment

For proper operation of this wireless thermometer, reference current (to pin 4 of DAC0808 - IC22) of the receiver unit should be pre-adjusted. To do this, follow the steps below:

Connect a known voltage source (not exceeding +5V) to any input of the ADC, say, at IN6 (pin 4) of the ADC. Switch on the transmitter unit. Connect a DMM across R26 of the receiver unit. Set the range switch to

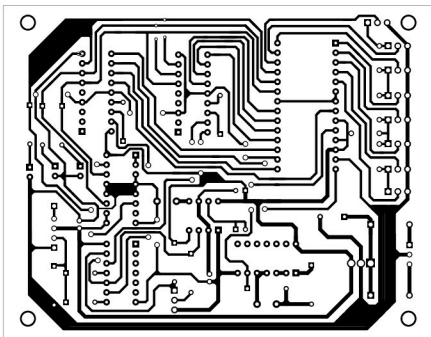


Fig. 4: An actual-size PCB pattern of the transmitter unit

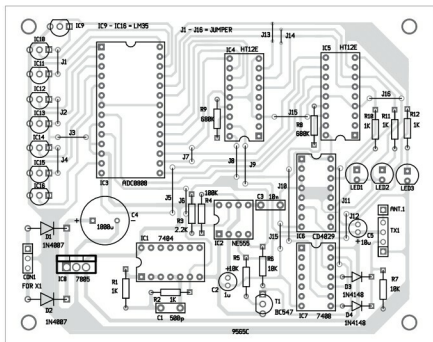


Fig. 5: Component layout of the PCB shown in Fig. 4

DC 200mV range, positive lead to ground and negative lead to top of R26. Switch on the receiver unit. LEDs at decoder outputs should start glowing to indicate the received voltage data. If source voltage is 1.5V, status of LEDs should be as listed in Table I.

So, received voltage = $(D \times 5) / 256 = (76 \times 5) / 256 = 1.50$ where D is the weight of the binary numbers represented by LED7 through LED14.

Now, adjust trim potmeter VR1 to get 150.00mV on the dial of the multi-meter. Connect another voltage source